

Efficacy and Safety of Sitagliptin, Metformin and Glimepiride FDC in Indian patients with T2DM:
A subgroup analysis of Phase 3 study based on BMI

Abstract number: A-1-135162-RSSDI2023

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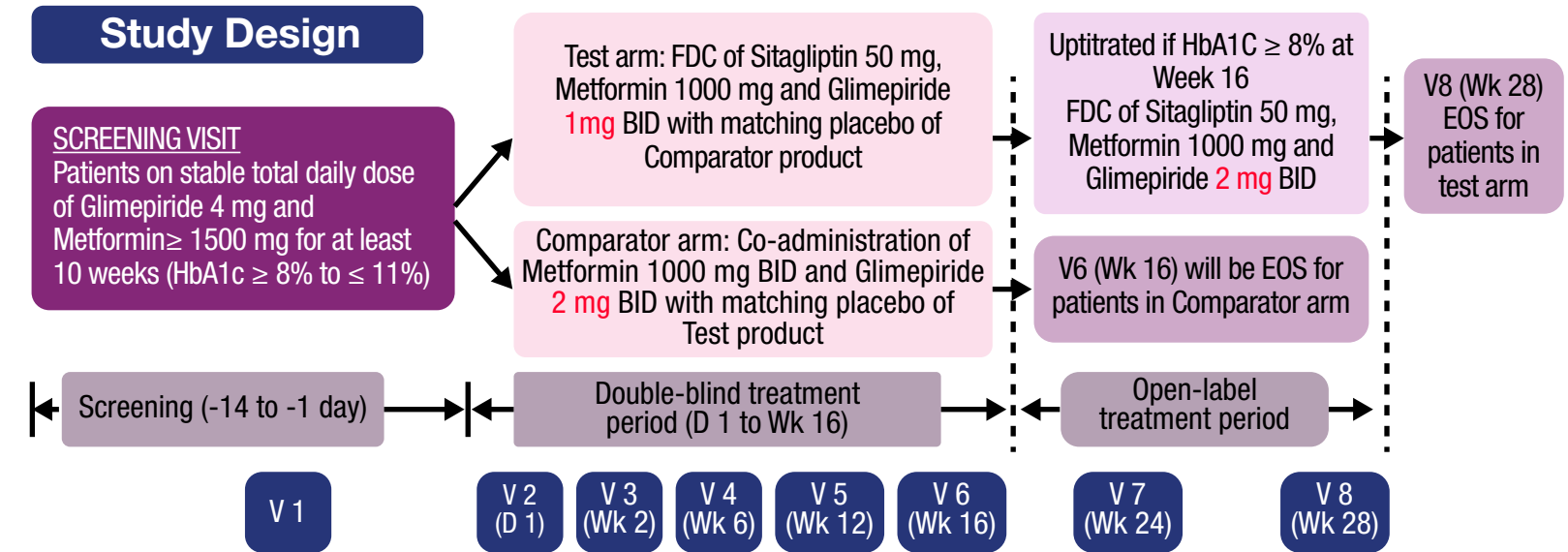
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BACKGROUND

- T2DM occurs at younger age and at lower levels of BMI in Indians compared with Caucasians. However, generalized obesity defined as a BMI of 25 kg/m² was observed in 28.6% of patients in ICMR-INDIAB-17 study, depicting a varied presentation of T2DM in India.¹
- Use of FDC of DPP-4 inhibitors, sulfonylurea, and Metformin may improve treatment compliance and provide superior glycemic control, leading to better patient outcomes.²
- We present here subgroup analysis of patients categorized by Baseline BMI at 16 Weeks from a phase III randomized, double-blind study (N=392).

METHODS

- This is a subgroup analysis of Phase 3 study (CTRI/2021/11/038169) of Sitagliptin+Glimepiride+Metformin FDC (50/1/1000 mg) in patients with T2DM (n=392).
- This study included patients with HbA1c ≥8% and ≤11% and BMI ≤45kg/m² and patients received either FDC Sitagliptin+Glimepiride+Metformin (50/1/1000 mg) BID (Test) or co-administration of Metformin 1000 mg BID and Glimepiride 2 mg BID (Comparator) for 16 Weeks.



- In this subgroup analysis, patients were categorized into subgroups (S1, S2, S3) based on Baseline BMI (S1: BMI <23 kg/m²; S2: BMI ≥23 - <25 kg/m²; S3: BMI ≥25 kg/m²).

RESULTS

Baseline Demographics

- A total of 392 patients were randomized (Test [n=190]; Comparator [n=202]).
- Baseline characteristics were comparable between Test arms and Comparator arms in all three subgroups.

	Subgroup 1: BMI <23 kg/m ²		Subgroup 2: BMI ≥23 - <25 kg/m ²		Subgroup 3: BMI ≥25 kg/m ²	
	Test N=33	Comparator N=36	Test N=61	Comparator N=54	Test N=96	Comparator N=112
Gender n (%)						
Male	16 (48.5%)	25 (69.4%)	32 (52.5%)	33 (61.1%)	54 (56.3%)	63 (56.3%)
Female	17 (51.5%)	11 (30.6%)	29 (47.5%)	21 (38.9%)	42 (43.8%)	49 (43.8%)
Mean ± Standard Deviation						
Age (years)	47.27± 8.46	47.92±10.03	46.62±8.88	45.74±8.58	48.28±8.43	47.34±9.17
Height (cm)	165.5±9.06	169.0±8.53	163.1±7.32	164.9±8.05	161.7±7.22	161.1±8.48
Weight (kg)	56.31±6.35	60.22±6.67	63.80±5.69	65.23±6.68	73.73±8.42	73.92±10.09
Body Mass Index (kg/m ²)	20.58±1.99	21.12±1.99	23.94±0.50	23.92±0.62	28.17±2.71	28.48±3.10
Glycated hemoglobin (%)	9.38±0.89	9.39±0.84	9.13±0.68	9.19±0.75	9.14±0.74	9.03±0.69
FBG (mg/dL)	178.2±49.91	178.4±44.57	181.7±33.50	178.5±39.04	173.9±36.70	188.3±39.29
PPBG (mg/dL)	276.3±77.44	249.2±66.99	255.9±51.49	252.5±54.96	244.0±44.85	255.5±63.80

Test: FDC of Sitagliptin, Metformin and Glimepiride tablets (50 mg/1000 mg/1 mg); Comparator: Co-administration of Metformin 1000 mg and Glimepiride tablets 2 mg.

Efficacy Results

Adjusted Mean Change in HbA1c (%) from Baseline

In all subgroups of BMI adjusted mean change (SE) in HbA1c (%) was statistically significant in the Test arm versus Comparator arm at Week 16.

Adjusted Mean Change in HbA1c (%) at Week 16

Subgroup 1 (BMI <23kg/m²)

Subgroup 2 (BMI ≥23 - <25kg/m²)

Subgroup 3 (BMI ≥25kg/m²)

Adjusted Mean HbA1c (%)

Test

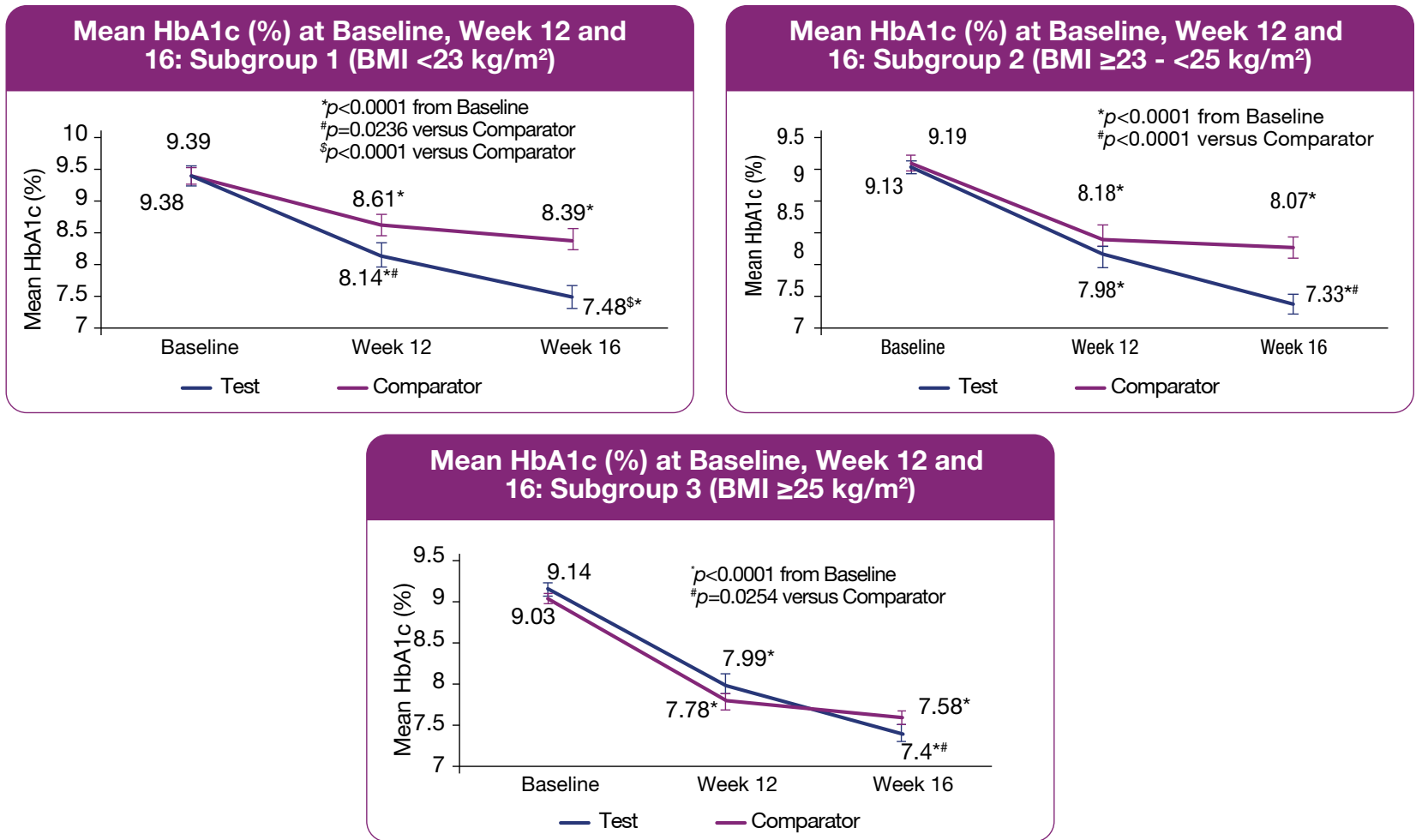
Comparator

*p<0.0001 versus Comparator

*p=0.0254 versus Comparator

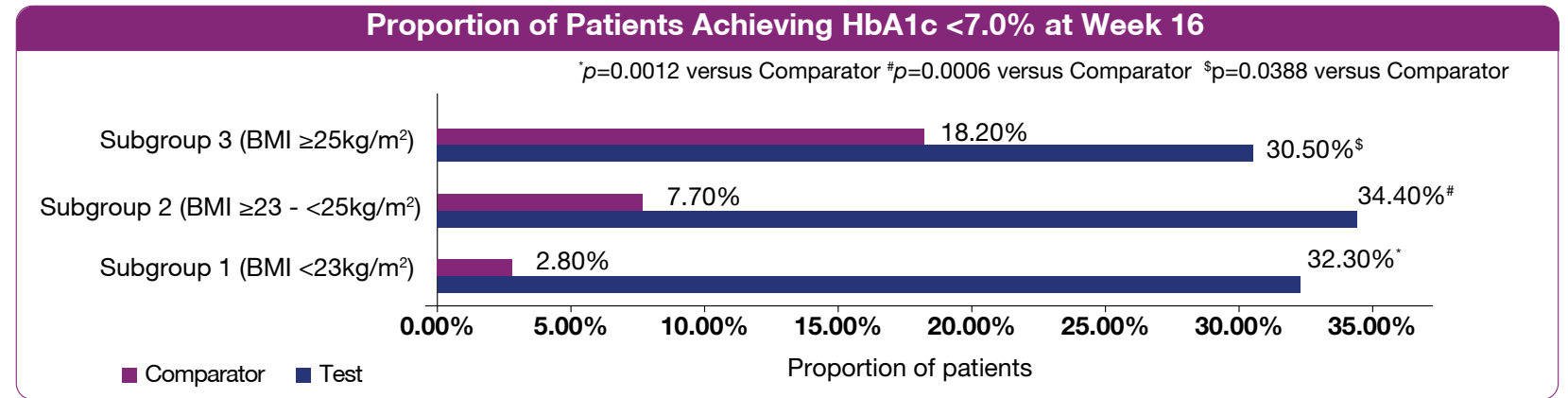
Mean HbA1c (%) at Baseline, Weeks 12 and 16

Both treatments in all three subgroups demonstrated significant reduction in HbA1c at Weeks 12 and 16 (p<0.0001 each) from Baseline.



Proportion of Patients Achieving HbA1C <7% at Week 16

In Test arm, significantly more patients achieved HbA1c <7.0% compared to Comparator at Week 16.



Fasting and Postprandial Blood Glucose at Week 6, 12, and 16

A statistically significant improvement in FBG and PPBG was observed from Baseline to Weeks 6, 12 and 16 in both the treatment arms in all three subgroups (p<0.0001 each).

Safety Results

Adverse Events: Overall Summary During the 16 Week Double Blind Treatment Period

- Safety profile was comparable between both arms in all three subgroups, and the treatments were safe and well-tolerated.
- The most commonly observed TEAEs were pyrexia (n=3), nausea (n=2) and vomiting (n=2) in S1; diarrhea (n=4), upper abdominal pain (n=3), and vomiting and headache (n=2, each) in S2; and headache (n=6), and vomiting, nausea, pyrexia and hypoglycemia (n=4, each) in S3.
- Overall 11 TEAEs (5 in Test and 6 in Comparator) were reported in 10 (14.5%) patients in S1; 19 TEAEs (8 in Test and 11 in Comparator) were reported in 16 (13.9%) patients in S2; and 37 TEAEs (18 in Test and 19 in Comparator) were reported in 29 (13.9%) patients in S3. One severe TEAE was reported in Comparator arm in S3.
- In S1, 5 TEAEs (3 in Test and 2 in Comparator arm) were reported as ‘unlikely’ related, 5 TEAEs (2 in Test and 3 in Comparator arm) were reported as ‘probable/likely’ related, and 1 TEAE (in Comparator arm) was reported as ‘possible’ related to the study treatment by Investigator.
- In S2, 11 TEAEs (4 in Test and 7 in Comparator arm) were reported as ‘unlikely’ related, 7 TEAEs (3 in Test and 4 in Comparator arm) were reported as ‘probable/likely’ related, and 1 TEAE (in Test arm) was reported as ‘possible’ related to the study treatment by Investigator.
- In S3, 23 TEAEs (12 in Test and 11 in Comparator arm) were reported as ‘unlikely’ related, 13 TEAEs (6 in Test and 7 in Comparator arm) were reported as ‘probable/likely’ related, and 1 TEAE (in Comparator arm) was reported as ‘possible’ related to the study treatment by Investigator.
- Action taken was “dose not changed” or “not applicable” for all the TEAEs in the study.
- No serious AE reported and none of the patient required dose modification.
- The outcome of all reported TEAEs was resolved at the end of study.
- None of the patient required rescue medications during the study.

DISCUSSION AND CONCLUSION

- Irrespective of the BMI at baseline, all three subgroups demonstrated a statistically significant improvement in HbA1c in Test versus Comparator.
- Significantly more proportion of patients achieved target HbA1c of <7% with FDC of Sitagliptin+Glimepiride+Metformin at Week 16 compared to co-administration of Metformin and high dose Glimepiride.
- A statistically significant reduction in PPBG and FBG was observed from Baseline to end of Weeks 6, 12, and 16 in both the treatment arms in all three subgroups.
- Thus, the addition of Sitagliptin to Metformin and a lower dose of Glimepiride offers superior glycemic control as compared to the co-administration of higher dose of Glimepiride with Metformin.
- Safety profile was comparable between both the arms regardless of the baseline BMI value, and both the study treatment were safe and well-tolerated.

Abbreviations:
BID: Twice daily; BMI: Body mass index; D: Day; DPP-4: dipeptidyl peptidase 4; EOS: End of study; FBG : Fasting Blood Glucose; FDC: Fixed dose combination; HbA1c: Glycated hemoglobin; N: Total number of patients; n: number of patients; PPBG : Postprandial Blood Glucose; SD: Standard deviation; T2DM:Type 2 diabetes mellitus; TEAEs: Treatment-emergent adverse events; V: Visit; Wk: Week

References:
1. Anjana RM *et al. Lancet Diabetes Endocrinol* 2023; 11: 474–489.
2. Kalra S *et al. J Family Med Prim Care.* 2020 Nov; 9(11): 5450–5457.